



ADAMA SCIENCE AND TECHNOLOGY UNIVERSITY

School of Electrical Engineering and Computing

Department of Computer Science and Engineering

Course: programming languages

Title: Tebeblang Language specifications

Group members	ID	SECTION
1. Yeabsira Goitom	Ugr/31390/15	2
2. Biruk selemon	Ugr/30332/15	
3. Biruk tadele	Ugr/30326/15	
4. Beriso Dedefo	Ugr/30274/15	

Submitted to : Hiwot Habtamu

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TibebLang Language Specification

1. Introduction

TibebLang is a beginner-friendly educational programming language designed using Amharic keywords. The main goal is to help Ethiopian beginners learn programming concepts more naturally through familiar language expressions.

TibebLang is a source-to-source language that translates its code into Python. Python was chosen as the target language because of its simple syntax, readability, Unicode support, and ease of execution.

2. Language Design Language

Name: TibebLang

Meaning:

TibebLang is a beginner-friendly educational programming language inspired by the Amharic word “**Tibeb**”, meaning wisdom, skill, or expertise. The language is designed to simplify programming education by using intuitive Amharic-based keywords, helping learners build strong foundational programming knowledge.

3.Keywords and Meanings

Feature	Tibeb Keyword	Meaning
Variable assignment	አድርግ	assign
Print output	አሳይ	print/show
If condition	ከሆነ	if
Else condition	ካልሆነ	else
While loop	እስከ	while
Function	ተግባር	function
Return	መልስ	return

End block	ጨርስ	end
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4. Syntax Rules

4.1 Variables

TibebLang አድርግ

x = 5

Python Output

x = 5

4.2 Printing Output

TibebLang አሳይ x

Python Output

print(x)

4.3 Arithmetic Expressions

Supported operators: + - * /

Example አድርግ total = 5 +

3 * 2 Python Output

total = 5 + 3 * 2

5. Control Flow 5.1

If / Else TibebLang

ከሆነ x > 5

አሳይ x

ካልሆነ

አሳይ 0

ጨርስ

Python Output

```
if x > 5:  
    print(x) else:  
    print(0)
```

5.2 While Loop TibebLang

```
እስከ x < 5 አሳይ  
x አድርግ x = x +  
1 ጨርስ
```

Python Output

```
while x < 5:  
    print(x) x  
    = x + 1
```

6. Functions

6.1 Function Definition TibebLang

```
ተግባር add(a, b)  
መልስ a + b  
ጨርስ
```

Python Output

```
def add(a, b):  
    return a + b
```

7. Beginner-Friendly Features

7.1 Natural Language Style

TibebLang syntax is designed to resemble natural Amharic sentence structure, making it easier for beginners to understand programming logic.

Example: ከሆነ

```
x > 5
```

This reads as: *“If x is greater than 5”*

7.2 Reduced Complexity

The language avoids complex symbols and strict syntax rules, reducing confusion for beginners and improving readability.

7.3 Local Language Integration

Using Amharic keywords allows learners to understand programming concepts in a familiar linguistic context, improving accessibility and learning speed.

8. System Architecture

TibebLang uses a parser-based translation system:

Source Code (.yl)



Lexer



Tokens



Parser



AST (Abstract Syntax Tree)



Code Generator



Python Output (.py)

Implementation Components:

- Lexical analysis
- Syntax parsing
- AST generation
- Python code generation

9. Unicode Support

TibebLang fully supports Amharic characters using UTF-8 encoding, ensuring correct processing of keywords and source code.

Example:

```
open(file, encoding="utf-8")
```

10. Conclusion

TibebLang demonstrates how local language integration can simplify programming education while still applying core compiler design concepts such as lexical analysis, parsing, AST construction, and code generation. It provides a bridge between natural language understanding and programming fundamentals for beginners.